

Introduction to Probability

Why study probability and statistics

You have a basket of 1000 apples and bananas and the task is to identify them based on their characteristics.

For the present discussion, let us focus on the color

An apple is **more** red than bananas

Pick up the 1000 apples and measure the value of redness for each apple

Experiment :: Picking an apple and measuring the red color value

Remember this experiment is **repeated** for each apple

Outcomes :: 1000 different readings. There is uncertainty in their values prior to measurement

Repeat the same experiment on the 1000 bananas and measure the value of redness for each banana

Outcomes :: 1000 different readings. There is uncertainty in their values prior to measurement

However, there is an **underlying pattern** that can be inferred

The red-ness values obtained for bananas are **likely** / **probable** to be **less** compared to the apples.

How do we quantify the terms -- 'more', 'less', 'likely' and 'probable'?

This course will help you to do that !



Introduction



PROBABILITY THEORY

Study of random phenomena.

Idea of repeated experiments yielding different outcomes that have certain underlying patterns about them.

When an experiment is performed, certain elementary events occur in different but *completely uncertain* ways.

An event typically relates to a set of outcomes that arise from an experiment.

<http://www.mhhe.com/engcs/electrical/papoulis/>

Random Experiment : An experiment is one whose outcome cannot be predicted precisely.

One out of a number of outcomes is possible in a random experiment.

A single performance of the random experiment is called a trial.

Classical Definition:

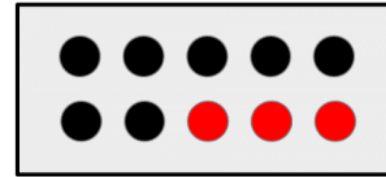
The probability of an event A is defined a-priori without actual experimentation as

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of possible outcomes}}$$

provided all these outcomes are *equally likely*.

Consider a box with 7 black and 3 red balls. In this case, there are two elementary outcomes: black ball or red ball.

$$\text{Probability of "selecting a black ball"} = \frac{7}{10}.$$



A fair die is rolled once. What is the probability of getting a '6' ?

$$\text{Probability of "getting a 6"} = \frac{1}{6}$$



Relative Frequency Definition

The probability of an event A is defined as

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n}$$

where

n_A is the number of occurrences of A

n is the total number of trials.

Determine the probability that a given number is divisible by a prime number p

Among the integers $1, 2, 3, \dots, n$, the numbers $p, 2p, \dots$ are divisible by p

$P\{a \text{ given number } N \text{ is divisible by a prime } p\}$

$$= \lim_{n \rightarrow \infty} \frac{n/p}{n} = \frac{1}{p}.$$

Sample space

Consider the experiment where two coins are simultaneously tossed.

The various elementary events are

$$\xi_1 = (H, H), \quad \xi_2 = (H, T), \quad \xi_3 = (T, H), \quad \xi_4 = (T, T)$$



$$\Omega = \{ \xi_1, \xi_2, \xi_3, \xi_4 \} \quad \text{Sample space}$$

The subset $A = \{ \xi_1, \xi_2, \xi_3 \}$ is the same as

“Head has occurred at least once” and qualifies as an event.

- The sample space is the collection of all possible outcomes of a random experiment.
- The elements are called *sample points*.
- A sample space may be *finite* , *countably infinite* or *uncountable*.
- Finite or countably infinite sample space is a *discrete sample space*.
- Uncountable sample space is a *continuous sample space*

Tossing a fair coin

The associated sample space is

Finite Discrete
sample space

$$\Omega = \{H, T\}$$



Throwing a fair die:

The associated sample space is

Finite
Discrete
sample space

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Let A = Event of Getting an even number

$$A = \{2, 4, 6\}$$



Tossing a fair coin until a head is obtained

The coin may be tossed any number of times before a head is obtained. Thus the possible outcomes are :

$$\Omega = \{ H, TH, TTH, TTTH, \dots \}$$

Countably infinite
Discrete sample space

The outcomes are countable but infinite in number.

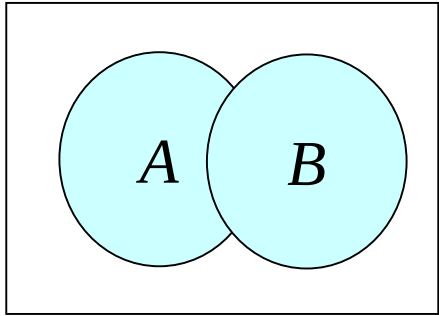
Picking a real number at random between -1 and +1

$$\Omega = \{ s | s \in R, -1 \leq s \leq 1 \}$$

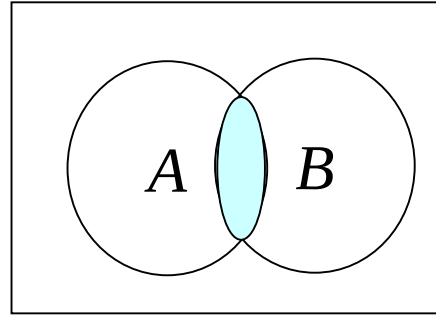
Continuous sample space.

The outcomes are uncountable and infinite in number.

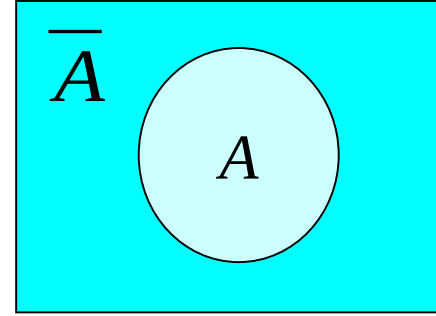
Set Operations



$$A \cup B$$



$$A \cap B$$



$$\overline{A}$$

Identity properties

$$A \cup A = A$$

$$A \cap A = A$$

$$A \cup \Omega = \Omega$$

$$A \cap \Omega = A$$

$$A \cup \phi = A$$

$$A \cap \phi = \phi$$

Commutative properties

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative properties

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive properties

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Complementary properties

$$A \cup \bar{A} = \Omega$$

$$A \cap \bar{A} = \phi$$

De Morgan's law

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

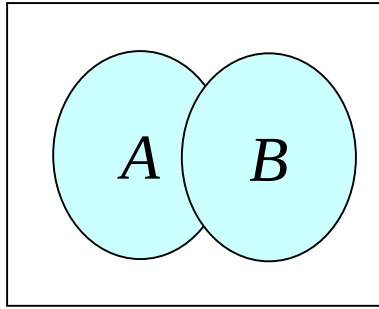
$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$



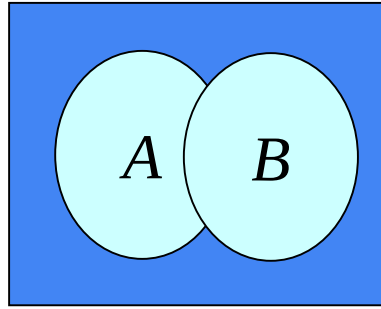
De Morgan's law



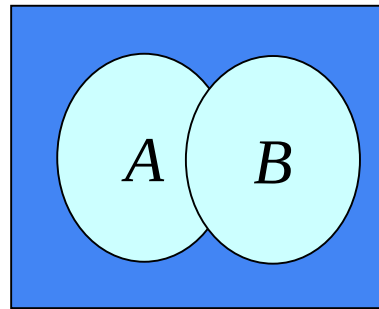
$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$



$A \cup B$



$\overline{A \cup B}$

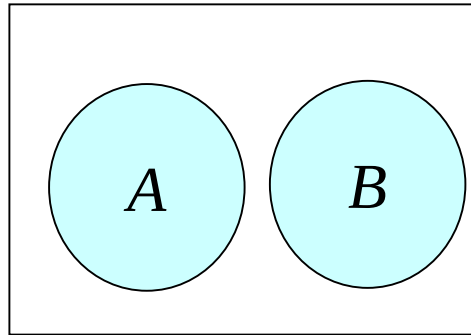


$\bar{A} \cap \bar{B}$

Thus the sets $A \cup B, A \cap B, \bar{A}, \bar{B}$ also qualify as events

Mutually exclusive sets

If $A \cap B = \phi$, then A and B are said to be mutually exclusive



$$A \cap B = \phi$$

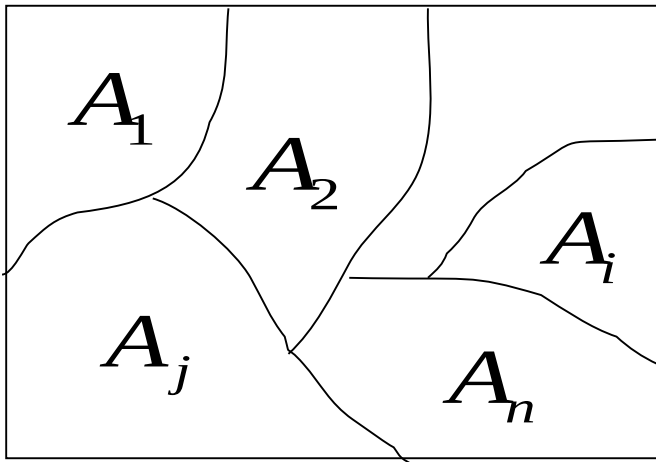
Partitioning of a set

A partition of Ω is a collection of mutually exclusive

subsets such that their union is Ω

$$A_i \cap A_j = \emptyset$$

$$\bigcup_{i=1} A_i = \Omega.$$



Sample space $\Omega = \{ \xi_1, \xi_2, \dots, \xi_k, \dots \}$

Ω has subsets $A, B, C, \dots$.

If A is a subset of Ω then $\xi \in \Omega$ implies $\xi \in A$

From A and B , we can generate other related subsets

$$A \cup B, A \cap B, \bar{A}, \bar{B},$$

$$A \cup B = \{ \xi \mid \xi \in A \text{ or } \xi \in B \}$$

$$A \cap B = \{ \xi \mid \xi \in A \text{ and } \xi \in B \}$$

$$\bar{A} = \{ \xi \mid \xi \notin A \}$$

Axioms of Probability

For any event A , we assign a number $P(A)$, called the probability of the event A . This number satisfies the following three conditions that act the axioms of probability.

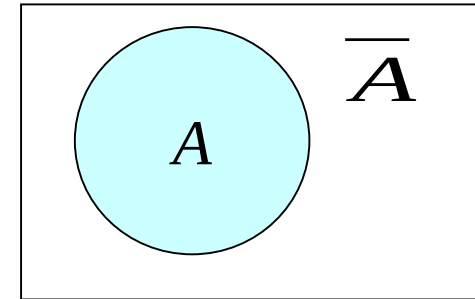
- (i) $P(A) \geq 0$ (Probability is a non - negative number)
- (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
- (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

Properties of Probability

We know

$$A \cup \bar{A} = \Omega$$

$$P(A \cup \bar{A}) = P(\Omega) = 1$$



$$A \cap \bar{A} \in \phi \quad (\text{Mutually exclusive events})$$

$$P(A \cup \bar{A}) = P(A) + P(\bar{A}) = 1$$

$$P(\bar{A}) = 1 - P(A).$$

Properties of Probability

$$A \cap \{\phi\} = \{\phi\}. \quad (\text{Mutually exclusive events})$$

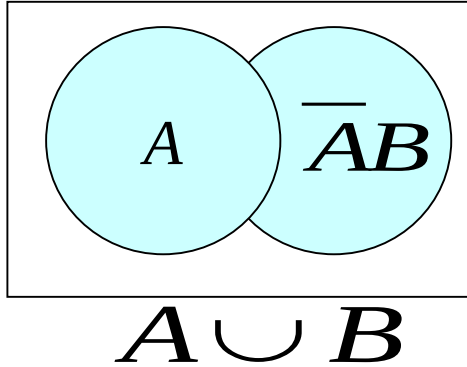
$$A \cup \{\phi\} = A,$$

$$P(A \cup \{\phi\}) = P(A) + P(\phi).$$

$$P(A) = P(A) + P(\phi).$$

$$P\{\phi\} = 0.$$

Properties of Probability



$$\overline{A}B = \overline{A} \cap B,$$

$$A \cup B = A \cup \overline{A}B,$$

$$P(A \cup B) = P(A \cup \overline{A}B). \quad (\text{Mutually exclusive events})$$

$$= P(A) + P(\overline{A}B)$$



$$\begin{aligned} B &= B \cap \Omega \\ &= B \cap (A \cup \bar{A}) \\ &= (B \cap A) \cup (B \cap \bar{A}) \\ &= BA \cup B\bar{A} \end{aligned}$$

$$P(B) = P(BA) + P(B\bar{A}), \quad (\text{Mutually exclusive events})$$

$$P(\bar{A}B) = P(B) - P(AB)$$

$P(A \cup B) = P(A) + P(B) - P(AB)$

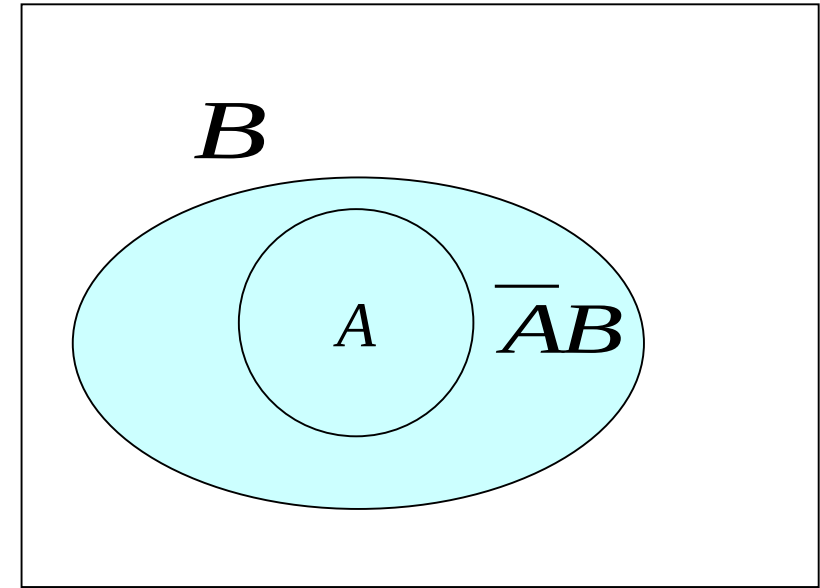
Properties of Probability

If $A \subset B$

$$B = A \cup B\bar{A}$$

$$P(B) = P(A) + P(B\bar{A}),$$

$$P(A) < P(B)$$



Axioms of Probability

For any event A , we assign a number $P(A)$, called the probability of the event A . This number satisfies the following three conditions that act the axioms of probability.

- (i) $P(A) \geq 0$ (Probability is a nonnegative number)
- (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
- (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

An integer is selected between 1 and 10. Assuming each number to be chosen as equally likely, Consider 2 events $A = \{\text{Select a odd number}\}$; $B = \{\text{Select a prime number}\}$

- 1) Calculate probability of choosing a prime number that is not odd
- 2) Calculate probability of choosing a number that is not prime and odd

$$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{2, 3, 5, 7\}$$

$$A \cap B = \{3, 5, 7\}$$

$$A \cup B = \{1, 2, 3, 5, 7, 9\}$$

- 1) probability of choosing a prime number that is not odd

$$\overline{A} \cap B = \{2\}$$

- 2) probability of choosing a number that is not prime and odd

$$\overline{A \cap B} = \{4, 6, 8, 10\}$$

An integer is selected between 1 and 40. Assuming each number to be chosen as equally likely, Calculate probability of

- 1) selecting a number that is divisible by 4 and 5.
- 2) Calculate probability of choosing a number that is divisible by 5 and not by 4.

Consider 2 events $A = \{\text{Number divisible by 4}\}$; $B = \{\text{Number divisible by 5}\}$

$$\Omega = \{1, 2, 3, 4, 5, \dots, 35, 36, 37, 38, 39, 40\}$$

$$A = \{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$$

$$B = \{5, 10, 15, 20, 25, 30, 35, 40\}$$

$$A \cap B = \{20, 40\}$$

$$A \cup B = \{4, 5, 8, 10, 12, 15, 16, 20, 24, 25, 28, 30, 32, 35, 36, 40\}$$

- 1) Probability of selecting a number that is divisible by 4 and 5.

$$A \cap B = \{20, 40\}$$

2) $P(A \cap B) = 0.05$

Probability of choosing a number that is divisible by 5 and not by 4.

$$\overline{A} \cap B = \{5, 10, 15, 25, 30, 35\}$$

$$P(\overline{A} \cap B) = 0.15$$

Consider tossing a fair coin until a head is obtained.

What is the probability of obtaining the head before 4 th toss

$$A_1 = \{\text{head appears for the 1st time on the first toss}\} \\ = \{h\}$$

$$A_2 = \{\text{head appears for the 1st time on the second toss}\} \\ = \{th\}$$

$$A_3 = \{\text{head appears for the 1st time on the third toss}\} \\ = \{tth\}$$

$$P(A_1) = P(h) = \frac{1}{2}$$

$$P(A_2) = P(th) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(A_3) = P(tth) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Each of the 3 events are pair wise mutually exclusive

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$$

$$P(A_1 \cup A_2 \cup A_3) = 0.875$$

Axioms of Probability

For any event A , we assign a number $P(A)$, called the probability of the event A . This number satisfies the following three conditions that act the axioms of probability.

- (i) $P(A) \geq 0$ (Probability is a nonnegative number)
- (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
- (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

In a coin tossing experiment,
where the same coin is tossed indefinitely

Let A = “head eventually appears”.

$$\begin{aligned} A_n &= \{\text{head appears for the 1st time on the } n\text{th toss}\} \\ &= \{\underbrace{t, t, t, \dots, t}_{n-1}, h\} \end{aligned}$$

$$A_i \cap A_j = \phi.$$

$$A = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_i \cup \dots.$$

If A_i are pair wise mutually exclusive, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

$$A_n = \{\text{head appears for the 1st time on the } n\text{th toss}\}$$
$$= \{\underbrace{t, t, t, \dots, t}_{n-1}, h\}$$

$$P(A_1) = \frac{1}{2}$$

$$P(A_2) = \frac{1}{4}$$

$$P(A_n) = \frac{1}{2^n}$$

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= \sum_{n=1}^{\infty} \frac{1}{2^n} \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \\ &= \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right) \\ &= 1 \end{aligned}$$